High-Filling Bose–Hubbard Models on Graphs: Uniform Rotor Comparison, Infrared-Regular Normal Forms, and a Fourth-Order Obstruction

Jizheng Chen[1,*] Tianrun Yu[2,*] Chaoyu Hu[3,†]

[1]Department of Mathematics, Chang'an University, Xi'an, China
[2]College of Control Science and Engineering, Zhejiang University, Hangzhou, China
[3]School of Automation, Hangzhou Dianzi University, Hangzhou, China

[*]These authors contributed equally.
[†]Corresponding author: evaristebernhardwiener@gmail.com

Abstract

The high-filling Bose–Hubbard model is commonly reduced to a quantum rotor by replacing the bosonic amplitude by its mean value. We establish a finite-graph version of this reduction that keeps the physical one-sided number space and does not impose an a priori cutoff on number fluctuations. The resulting quadratic-form comparison is uniform in graph size and controls number moments, bad-number tails, and the error made when the normalized bosonic annihilation operator is replaced by the phase shift. On a symmetric finite number window we derive the complete second-order normal form. The particle–hole odd first-order term admits an exact completion of squares and satisfies a graph-uniform resolvent estimate, while the direct second-order term is a total double commutator plus a phase-force square. The induced second-order phase potential has an explicit neutral-character expansion and can be compared, through Ginibre monotonicity, with a standard rotor of slightly reduced stiffness. Finally, a three-site classical principal-symbol calculation shows that an all-order ferromagnetic closure fails at fourth order. These results rigorously delimit both the validity and the structural limitations of the high-filling rotor approximation. No thermodynamic Bose–Einstein condensation statement is claimed.

Keywords: Bose–Hubbard model; quantum rotor; high filling; operator comparison; infrared estimate; double commutator; Griffiths–Ginibre inequality.

1 Introduction

The Bose–Hubbard Hamiltonian is a standard lattice model for interacting bosons and for the superfluid–Mott transition [3]. At a large integer filling $\bar{n}$, one writes the local occupation as $\bar{n} + M_x$ and heuristically replaces the normalized annihilation operator by a phase shift. This leads to the compact $U(1)$ quantum rotor. The reduction is widely used in effective-action calculations and in the analysis of Josephson-junction-type systems; see, for example, [8, 9]. Its rigorous content, however, depends on what is being compared: energies, quadratic forms, correlation functions, or low-energy spectral data.

The purpose of this paper is deliberately narrower than a derivation of thermodynamic long-range order. We isolate a package of finite-graph statements that can be proved without assuming condensation and without introducing a number cutoff in the basic comparison. We then compute the finite-window normal form through second order and identify algebraic mechanisms that prevent the most obvious infrared losses. The same calculation also exhibits a sharp limitation: the generalized ferromagnetic structure visible at second order does not persist to all orders.

Our main results are as follows.

1. On the physical half-line $M_x \geq -\bar{n}$, the amplitude defect $A_x - S_x$ is bounded pointwise by $|M_x|/\bar{n}$. This yields a graph-uniform lower form bound by a quantum rotor with slightly weakened stiffness, together with number-moment and tail estimates.
2. On a symmetric finite number window, the hopping term has a complete second-order expansion. The first-order particle–hole odd term is an anticommutator with a real phase field; it can be completed into a magnetic square and satisfies a resolvent estimate whose constant is independent of graph volume and of the smallest nonzero graph momentum.
3. The direct second-order term is a total double commutator plus the square of the phase force. After combining it with the square-completion correction, the remaining phase potential is relatively bounded by the original rotor potential.
4. On regular graphs, the second-order phase potential has a neutral-character expansion with nonnegative couplings. Standard Ginibre–Griffiths monotonicity therefore compares it with a standard rotor having renormalized stiffness.
5. A three-site classical elimination shows that at order $\bar{n}^{-4}$ a character with the opposite sign appears. Thus a naive all-order ferromagnetic expansion is already false at the principal-symbol level.

The work does *not* prove convergence of all correlation functions from the Bose–Hubbard model to the rotor, a volume-uniform Gaussian domination estimate for the full amplitude model, or a new Bose–Einstein condensation regime. These limitations are part of the mathematical statement rather than technical qualifications hidden in the proofs.

Relation to earlier work. The high-filling quantum-rotor description of the Bose–Hubbard model is standard in the physics literature [8, 9]. Our contribution is not the heuristic mapping itself, but the finite-graph operator estimates and the explicit protected normal form. The correlation comparison in Section 7 uses the general Ginibre framework [5, 4, 7] and related quantum-rotor positivity methods; it is quoted as a standard input rather than claimed as a new correlation inequality. The estimates are also conceptually distinct from asymptotic localization results such as [2].

2 Models and notation

Let $G = (V, E)$ be a finite undirected graph of maximum degree Δ. When regularity is needed, the common degree is denoted by q. Each undirected edge is assigned an arbitrary orientation $e = (x, y)$; all final expressions are orientation independent.

The compact rotor at one site acts on $\ell^2(\mathbb{Z})$. We use the number operator

$$M|m\rangle = m|m\rangle$$

and the bilateral shift

$$S|m\rangle = |m-1\rangle.$$

On an edge $e = (x, y)$, set

$$T_e = S_x^\dagger S_y, \qquad \mathcal{A}_e = 2 - T_e - T_e^\dagger = 2 - 2\cos(\theta_x - \theta_y). \tag{1}$$

The rotor Hamiltonian is

$$H_0 = H_c + V, \qquad H_c = \frac{U}{2}\sum_x M_x^2, \qquad V = J\sum_e \mathcal{A}_e. \tag{2}$$

For the physical high-filling block model, the local number deviation satisfies $M_x \geq -\bar{n}$. The lowering shift is the partial isometry

$$S|-\bar{n}\rangle = 0, \qquad S|m\rangle = |m-1\rangle \quad (m > -\bar{n}).$$

Define

$$A_x = S_x\sqrt{1 + M_x/\bar{n}}, \tag{3}$$

and

$$H_{\bar{n}} = \frac{U}{2}\sum_x M_x^2 + J\sum_{e=(x,y)} (A_x - A_y)^\dagger(A_x - A_y). \tag{4}$$

On a q-regular graph and in the zero-total-deviation sector $\sum_x M_x = 0$, the onsite part coming from $\sum_e (A_x^\dagger A_x + A_y^\dagger A_y)$ is an additive constant. This is the sector in which the comparison with H_0 will be used.

3 A cutoff-free rotor comparison

Lemma 3.1 (Single-site amplitude defect)**.** *Let $D = A - S$ on the physical one-site Hilbert space. Then*

$$D^\dagger D \le \frac{M^2}{\bar{n}^2}. \tag{5}$$

Proof. Away from the bottom state,

$$D = S\big(\sqrt{1 + M/\bar{n}} - 1\big).$$

For $t \ge -1$,

$$(\sqrt{1+t} - 1)^2 = \frac{t^2}{(\sqrt{1+t}+1)^2} \le t^2.$$

At the bottom state the left-hand side vanishes, so the same spectral inequality holds everywhere. □

Theorem 3.2 (Uniform quadratic-form comparison)**.** *For every $0 < \kappa < 1$,*

$$H_{\bar{n}} \ge \left(\frac{U}{2} - \frac{2J\Delta}{\kappa\bar{n}^2}\right)\sum_x M_x^2 + J(1-\kappa)Q_{\mathrm{ph}}, \tag{6}$$

where

$$Q_{\mathrm{ph}} := \sum_{e=(x,y)} (S_x - S_y)^\dagger(S_x - S_y). \tag{7}$$

Proof. Write

$$A_x - A_y = (S_x - S_y) + (D_x - D_y).$$

The elementary inequality

$$\|u + v\|^2 \ge (1-\kappa)\|u\|^2 - \kappa^{-1}\|v\|^2$$

gives the phase term in (6). Moreover,

$$\sum_e (D_x - D_y)^\dagger(D_x - D_y) \le 2\Delta\sum_x D_x^\dagger D_x \le \frac{2\Delta}{\bar{n}^2}\sum_x M_x^2$$

by (5). Substitution into (4) proves the claim. □

A convenient consequence is obtained when $\bar{n}^2 \ge 8J\Delta/U$. Choosing

$$\kappa = \frac{8J\Delta}{U\bar{n}^2}$$

yields

$$H_{\bar{n}} \ge \frac{U}{4}\sum_x M_x^2 + J\left(1 - \frac{8J\Delta}{U\bar{n}^2}\right)Q_{\mathrm{ph}}. \tag{8}$$

Corollary 3.3 (Moments, tails, and observable replacement)**.** *Assume that G is q-regular, work in the sector $\sum_x M_x = 0$, and let Ψ be a ground state of $H_{\bar{n}}$. Under $\bar{n}^2 \geq 8J\Delta/U$,*

$$\frac{1}{|V|}\sum_x \langle M_x^2\rangle_\Psi \leq \frac{4\Delta J}{U}, \tag{9}$$

$$\frac{1}{|V|}\sum_x \langle (A_x - S_x)^\dagger (A_x - S_x)\rangle_\Psi \leq \frac{4\Delta J}{U\bar{n}^2}, \tag{10}$$

$$\frac{1}{|V|}\sum_x \langle \mathbf{1}_{\{|M_x|>K\}}\rangle_\Psi \leq \frac{4\Delta J}{UK^2}. \tag{11}$$

Proof. Use the product number state $M_x = 0$ as a trial state. Each edge contributes at most $2J$, hence

$$E_0 \leq 2J|E| \leq J\Delta|V|.$$

Equation (8) then gives (9). Equation (10) follows from (5), and (11) follows from

$$\mathbf{1}_{\{|M|>K\}} \leq M^2/K^2.$$

□

The estimates are uniform in $|V|$ and do not use a spectral gap. They are form comparisons; they do not by themselves imply convergence of all low-energy correlations to those of the compact rotor.

4 Finite-window normal form

Fix $K \geq 1$ and let P_K project onto

$$\mathrm{span}\{|m\rangle : -K \leq m \leq K\}.$$

Write $P_K^{(V)}$ for its tensor product over the graph. All identities in this section concern the compressed matrix elements

$$P_K^{(V)} H_{\bar{n}} P_K^{(V)}$$

of the original physical operators. We do not rebuild the Hamiltonian from $P_K A_x P_K$, because that different operation produces number-window boundary projectors. Assume $\bar{n} \geq 2(K+1)$, so the Taylor expansion is uniform on the window. For $e = (x, y)$, define

$$q_e = M_x + M_y, \qquad d_e = M_x - M_y, \qquad T_e = S_x^\dagger S_y. \tag{12}$$

Lemma 4.1 (Second-order hopping expansion)**.** *On the compressed two-site window,*

$$A_x^\dagger A_y = T_e\left[1 + \frac{q_e+1}{2\bar{n}} - \frac{(d_e+1)^2}{8\bar{n}^2}\right] + R_e, \tag{13}$$

with

$$\|R_e\| \leq C\frac{(K+1)^3}{\bar{n}^3}. \tag{14}$$

Proof. The nonzero matrix element of $P_K^{(V)} A_x^\dagger A_y P_K^{(V)}$ equals that of the compressed T_e multiplied by

$$\sqrt{\left(1 + \frac{M_x+1}{\bar{n}}\right)\left(1 + \frac{M_y}{\bar{n}}\right)}.$$

The claimed expression is the Taylor expansion of $\sqrt{(1+s)(1+t)}$ at $(0,0)$, whose linear part is $(s+t)/2$ and quadratic part is $-(s-t)^2/8$. The third derivatives are uniformly bounded for $|s|, |t| \leq 1/2$. Both sides vanish on transitions that leave the window. Thus the identity concerns the compressed product $A_x^\dagger A_y$, not a product of separately compressed amplitudes. □

Theorem 4.2 (Complete second-order Hamiltonian expansion)**.** *Let*

$$H_{\bar{n},K} := P_K^{(V)} H_{\bar{n}} P_K^{(V)}.$$

If G is q-regular and we restrict to $\sum_x M_x = 0$, then, up to an additive constant,

$$H_{\bar{n},K} = H_{0,K} + \bar{n}^{-1} H_1 + \bar{n}^{-2} H_2 + R_3, \tag{15}$$

where

$$H_1 = \frac{1}{2} K_J + D_{\mathrm{odd}}, \qquad K_J = -J \sum_e (T_e + T_e^\dagger), \tag{16}$$

$$D_{\mathrm{odd}} = -\frac{J}{2} \sum_e q_e (T_e + T_e^\dagger) = \frac{J}{2} \sum_e q_e \mathcal{A}_e, \tag{17}$$

$$H_2 = \frac{J}{8} \sum_e \left[T_e (d_e + 1)^2 + (d_e + 1)^2 T_e^\dagger \right], \tag{18}$$

$$\|R_3\| \le CJ|E| \frac{(K+1)^3}{\bar{n}^3}. \tag{19}$$

Proof. Compress (4) as a whole and insert (13) and its adjoint into the cross-hopping terms. The onsite identity

$$P_K^{(V)} A_x^\dagger A_x P_K^{(V)} = P_K^{(V)} \left(1 + \frac{M_x}{\bar{n}} \right) P_K^{(V)}$$

remains exact; this is why the Hamiltonian is not reconstructed from separately compressed amplitudes. Since q_e commutes with T_e and

$$\sum_e q_e = q \sum_x M_x = 0,$$

the first-order coefficient separates into $K_J/2$ and D_{odd}. The second-order coefficient is (18); summing the edge remainders gives (19). □

Proposition 4.3 (Particle–hole cancellation)**.** *On the symmetric window define $C|m\rangle = |-m\rangle$. Then*

$$C H_0 C^* = H_0, \qquad C K_J C^* = K_J, \qquad C D_{\mathrm{odd}} C^* = -D_{\mathrm{odd}}.$$

Consequently, for every particle–hole even source X and every $\beta > 0$,

$$\langle D_{\mathrm{odd}} \rangle_{\beta, H_0 - hX} = 0. \tag{20}$$

Proof. One has $CMC^* = -M$ and $CSC^* = S^\dagger$. Thus T_e is exchanged with $T_e^\dagger$, $\mathcal{A}_e$ is fixed, and q_e changes sign. The Gibbs density of $H_0 - hX$ commutes with C, so the trace of the odd operator D_{odd} vanishes. □

5 The odd term: completion of squares and a graph-uniform resolvent bound

Define the real multiplication operators

$$b_x = \frac{J}{2} \sum_{e \ni x} \mathcal{A}_e. \tag{21}$$

Proposition 5.1 (Phase-defect representation)**.** *In the q-regular zero-charge sector,*

$$D_{\mathrm{odd}} = \frac{1}{2} \sum_x \{ b_x, M_x \}. \tag{22}$$

Proof. For an edge $e = (x, y)$, the total edge charge $q_e = M_x + M_y$ commutes with $\mathcal{A}_e$. Summing the two incidence contributions gives

$$\frac{J}{4}\{\mathcal{A}_e, q_e\} = \frac{J}{2} q_e \mathcal{A}_e,$$

which is exactly the edge term in (17). □

Proposition 5.2 (Exact completion of squares)**.** *With $\varepsilon = \bar{n}^{-1}$,*

$$\frac{U}{2}\sum_x M_x^2 + \varepsilon D_{\text{odd}} = \frac{U}{2}\sum_x \left(M_x + \frac{\varepsilon b_x}{U}\right)^2 - \frac{\varepsilon^2}{2U}\sum_x b_x^2. \tag{23}$$

Proof. Expand each noncommutative square. The cross term is $\frac{\varepsilon}{2}\{M_x, b_x\}$; no commutativity between M_x and b_x is required. □

In the angle representation, (23) is a Schrödinger operator with a real magnetic vector potential. The standard diamagnetic inequality therefore compares its ground-state energy with the same scalar potential and zero vector potential. The statement remains valid in the zero-total-charge sector because taking the modulus preserves invariance under common angle shifts.

Theorem 5.3 (Composite infrared estimate)**.** *Let $\Omega > 0$ be the normalized ground state of H_0 in the zero-total-charge sector, set $K_0 = H_0 - E_0$, and let $K_{0,\perp}^{-1}$ denote the inverse on $\Omega^\perp$. For real edge coefficients u_e, define*

$$\mathcal{O}(u) = \sum_{e=(x,y)} u_e (M_x + M_y)\mathcal{A}_e. \tag{24}$$

Then

$$\left\langle \mathcal{O}(u)\Omega, K_{0,\perp}^{-1}\mathcal{O}(u)\Omega \right\rangle \le \frac{4\Delta}{U}\sum_e |u_e|^2 \langle \Omega, \mathcal{A}_e\Omega\rangle \le \frac{16\Delta}{U}\sum_e |u_e|^2. \tag{25}$$

Proof. Set

$$B_x(\theta) = \sum_{e \ni x} u_e \mathcal{A}_e(\theta).$$

Since each $\mathcal{A}_e$ depends only on $\theta_x - \theta_y$, the full divergence satisfies

$$\sum_x \partial_{\theta_x} B_x = 0.$$

Hence $\mathcal{O}(u) = -iB \cdot \nabla$. The ground-state transform gives

$$\langle f\Omega, K_0 f\Omega\rangle = \frac{U}{2}\int |\nabla f|^2 \Omega^2.$$

Integration by parts yields

$$\langle f\Omega, \mathcal{O}(u)\Omega\rangle = \frac{i}{2}\int \Omega^2 B \cdot \nabla \overline{f}.$$

Optimizing the resolvent variational formula gives an upper bound by $(2U)^{-1}\int |B|^2\Omega^2$. Finally,

$$|B|^2 \le 2\Delta \sum_e |u_e|^2 \mathcal{A}_e^2, \qquad \mathcal{A}_e^2 \le 4\mathcal{A}_e \le 16,$$

which proves (25). □

The important feature of (25) is the absence of the graph spectral gap. The composite density–phase operator has a built-in transport cancellation; estimating it only by operator norm would lose this structure.

6 The direct second-order term

Lemma 6.1 (Edge double commutator)**.** *For* $C_e = M_x^2 + M_y^2$,

$$T_e(d_e+1)^2 + (d_e+1)^2 T_e^\dagger = \frac{1}{4}[C_e, [C_e, T_e + T_e^\dagger]]. \tag{26}$$

Proof. The shift T_e changes d_e by two and changes C_e by $2(d_e+1)$. Thus

$$[C_e, T_e] = 2T_e(d_e+1),$$

and a second commutator gives $4T_e(d_e+1)^2$. Taking the adjoint gives the $T_e^\dagger$ term with the displayed ordering. □

Theorem 6.2 (Global normal form)**.** *The operator* H_2 *in* (18) *satisfies*

$$H_2 = -\frac{1}{8U^2}[H_c, [H_c, V]], \tag{27}$$

and

$$H_2 = -\frac{1}{8U^2}[H_0, [H_0, V]] + \frac{1}{8U}|\nabla_\theta V|^2. \tag{28}$$

Proof. Equation (27) follows by summing (26) and using

$$V = 2J|E| - J\sum_e (T_e + T_e^\dagger).$$

Since $[H_c, V] = [H_0, V]$,

$$[H_c, [H_c, V]] = [H_0, [H_0, V]] - [V, [H_0, V]].$$

For $H_c = -(U/2)\Delta_\theta$,

$$[V, [H_c, V]] = U|\nabla_\theta V|^2,$$

which proves (28). □

Combining the phase-force term in (28) with the negative scalar correction in (23) defines

$$W_{\mathrm{sc}} = \frac{1}{8U}|\nabla_\theta V|^2 - \frac{1}{2U}\sum_x b_x^2. \tag{29}$$

Proposition 6.3 (Relative phase bound)**.** *Pointwise on the phase torus,*

$$|W_{\mathrm{sc}}| \le \frac{J\Delta}{U} V. \tag{30}$$

Proof. Let

$$Q_x = \sum_{y\sim x}\left(e^{i(\theta_x - \theta_y)} - 1\right).$$

Then

$$\partial_{\theta_x} V = 2J\ \mathrm{Im}\, Q_x, \qquad b_x = -J\ \mathrm{Re}\, Q_x,$$

so

$$W_{\mathrm{sc}} = \frac{J^2}{2U}\sum_x \left[(\mathrm{Im}\, Q_x)^2 - (\mathrm{Re}\, Q_x)^2\right].$$

By Cauchy–Schwarz,

$$|Q_x|^2 \le \deg(x)\sum_{y\sim x}|e^{i(\theta_x-\theta_y)} - 1|^2 = \deg(x)\sum_{y\sim x}\mathcal{A}_{xy}.$$

Summing over x yields

$$\sum_x |Q_x|^2 \le \frac{2\Delta}{J} V,$$

which gives (30). □

Thus $\bar{n}^{-2} W_{\mathrm{sc}}$ is relatively of size $J/(U\bar{n}^2)$ compared with the principal phase potential. The statement is pointwise and volume independent.

7 Neutral-character structure at second order

Proposition 7.1 (Character expansion)**.** *If G is q-regular, then*

$$W_{\mathrm{sc}} = -\frac{J^2}{4U} \sum_x (Q_x^2 + \overline{Q}_x^2), \tag{31}$$

and

$$W_{\mathrm{sc}} = \frac{J^2 q}{U} \sum_x \sum_{y\sim x} \cos(\theta_x - \theta_y) - \frac{J^2}{2U} \sum_x \sum_{y,z\sim x} \cos(2\theta_x - \theta_y - \theta_z) + \text{const.} \tag{32}$$

Proof. The first formula follows from

$$(\operatorname{Im} Q)^2 - (\operatorname{Re} Q)^2 = -\frac{1}{2}(Q^2 + \overline{Q}^2).$$

Expanding Q_x^2 and taking real parts gives

$$Q_x^2 + \overline{Q}_x^2 = 2 \sum_{y,z\sim x} \cos(2\theta_x - \theta_y - \theta_z) - 4q \sum_{y\sim x} \cos(\theta_x - \theta_y) + 2q^2.$$

□

Consequently, up to an additive constant, the phase Hamiltonian through order $\bar{n}^{-2}$ can be written as

$$V_{\mathrm{eff}} = J_* \sum_e \mathcal{A}_e - \frac{J^2}{2U\bar{n}^2} \sum_x \sum_{y,z\sim x} \cos(2\theta_x - \theta_y - \theta_z), \tag{33}$$

where

$$J_* = J - \frac{qJ^2}{U\bar{n}^2}. \tag{34}$$

Every new character in (33) is neutral: its integer coefficients sum to zero. Its Hamiltonian coefficient has the ferromagnetic sign.

Proposition 7.2 (Standard Ginibre monotonicity for neutral characters)**.** *Consider a finite-volume quantum rotor*

$$H(\boldsymbol{J}) = \frac{U}{2} \sum_x M_x^2 - \sum_\gamma J_\gamma \cos(\gamma \cdot \theta), \qquad \sum_x \gamma_x = 0, \qquad J_\gamma \ge 0. \tag{35}$$

For neutral η, the finite-temperature expectation $\langle \cos(\eta \cdot \theta) \rangle$ is nondecreasing in every coupling J_γ. The same is true for zero-temperature limit states.

Standard input. A Trotter discretization in imaginary time has nonnegative Fourier coefficients both for the rotor heat kernel and for $\exp(tJ_\gamma \cos(\gamma \cdot \theta))$. Ginibre's positive-function cone gives the second correlation inequality for the resulting classical compact-abelian spin system [5, 4]. Passing to the Trotter and zero-temperature limits gives the statement; see also quantum formulations such as [7, 6, 1]. We use this as an external positivity theorem rather than as a new result. □

Corollary 7.3 (Comparison with a weakened standard rotor)**.** *If $J_* > 0$, then for every pair x, y,*

$$\langle \cos(\theta_x - \theta_y) \rangle_{V_{\text{eff}}} \geq \langle \cos(\theta_x - \theta_y) \rangle_{\text{standard rotor of stiffness } J_*}. \tag{36}$$

Proof. Start from the standard rotor with stiffness J_* and turn on the nonnegative neutral couplings in (33) one at a time. The preceding monotonicity statement yields (36). □

8 A fourth-order obstruction to all-order ferromagnetic closure

The preceding section concerns the second-order effective phase potential. It is natural, but incorrect, to conjecture that all higher-order density eliminations preserve the same sign structure. The following three-site principal-symbol calculation gives a strict counterexample.

Consider the path 0–1–2. Set $\theta_0 = 0$, $\theta_1 = a$, $\theta_2 = b$, $c_1 = \cos a$, and $c_2 = \cos(b - a)$. Under $m_0 + m_1 + m_2 = 0$, define

$$\mathcal{H}_\varepsilon = \frac{U}{2} \sum_{i=0}^{2} m_i^2 - K_0 \left[\sqrt{(1 + \varepsilon m_0)(1 + \varepsilon m_1)}\, c_1 + \sqrt{(1 + \varepsilon m_1)(1 + \varepsilon m_2)}\, c_2 \right]. \tag{37}$$

Theorem 8.1 (Three-site fourth-order obstruction)**.** *The local minimizer near $m = 0$ has*

$$m_0 = \frac{K_0(c_1 - 2c_2)}{6U}\varepsilon + \frac{K_0^2 c_1 c_2}{8U^2}\varepsilon^3 + O(\varepsilon^5), \tag{38}$$

$$m_1 = \frac{K_0(c_1 + c_2)}{6U}\varepsilon - \frac{K_0^2 c_1 c_2}{4U^2}\varepsilon^3 + O(\varepsilon^5), \tag{39}$$

and the minimized energy equals

$$\begin{aligned} E_{\text{eff}} = & -K_0(c_1 + c_2) + \frac{K_0^2 \varepsilon^2}{12U}(-c_1^2 + c_1 c_2 - c_2^2) \\ & + \frac{K_0^3 \varepsilon^4}{32U^2} c_1 c_2 (c_1 + c_2) + O(\varepsilon^6). \end{aligned} \tag{40}$$

The ε^4 term contains

$$+\frac{K_0^3 \varepsilon^4}{128U^2} \cos(a + b), \tag{41}$$

which has the opposite sign from a ferromagnetic Hamiltonian term $-\lambda \cos(a + b)$ with $\lambda \geq 0$.

Proof. Write $m_2 = -m_0 - m_1$ and solve the two stationary equations with the ansatz

$$m_0 = a_1 \varepsilon + a_3 \varepsilon^3 + O(\varepsilon^5), \qquad m_1 = b_1 \varepsilon + b_3 \varepsilon^3 + O(\varepsilon^5).$$

Coefficient matching gives (38)–(39). Substitution yields (40). Finally,

$$\begin{aligned} c_1 c_2 (c_1 + c_2) = & \frac{1}{2}\cos a + \frac{1}{2}\cos(a - b) + \frac{1}{4}\cos(a + b) + \frac{1}{4}\cos(a - 2b) \\ & + \frac{1}{4}\cos(3a - b) + \frac{1}{4}\cos(3a - 2b), \end{aligned}$$

which proves (41). □

The theorem is intentionally stated as a classical principal-symbol obstruction. It does not by itself identify the complete fourth-order quantum effective Hamiltonian. It does rule out arguments asserting that the density-amplitude expansion is sign-preserving at every order for purely algebraic reasons.

Lemma 8.2 (Neutral characters are controlled by local gradients)**.** *Let* $\gamma \in \mathbb{Z}^V$ *satisfy* $\sum_x \gamma_x = 0$ *and suppose* $\operatorname{supp}\gamma$ *lies in a connected finite subgraph* X*. Then*

$$1 - \cos(\gamma \cdot \theta) \le C_{\gamma,X} \sum_{e \subset X} \big[1 - \cos(\nabla_e \theta)\big]. \tag{42}$$

Proof. Choose an integer flow c_e on a spanning tree of X with $\operatorname{div} c = \gamma$. Then

$$\gamma \cdot \theta = \sum_e c_e \nabla_e \theta.$$

The inequalities

$$|e^{in\phi} - 1| \le |n|\,|e^{i\phi} - 1|$$

and a telescoping product bound give

$$|e^{i\gamma\cdot\theta} - 1| \le \sum_e |c_e|\,|e^{i\nabla_e\theta} - 1|.$$

Squaring and applying Cauchy–Schwarz proves (42). □

9 Scope, limitations, and possible extensions

The results establish a rigorous and graph-uniform structural approximation, but they stop short of a complete low-energy equivalence. Three distinctions are essential.

- The cutoff-free comparison controls quadratic forms, moments, and tails. It does not imply convergence of the full spectrum or of arbitrary correlation functions.
- The second-order character comparison applies to the effective compact-rotor Hamiltonian. It is not a Gaussian domination theorem for the original amplitude model.
- The fourth-order obstruction concerns the classical principal symbol. It invalidates an all-order sign conjecture but does not preclude perturbative control by other mechanisms.

A stronger theorem would require one of two additional ingredients: either a volume-uniform source estimate that transfers rotor infrared bounds to the amplitude model, or a direct norm-resolvent/low-energy-subspace comparison in a parameter regime with controlled number tails. Neither statement follows from the present normal form alone.

The package nevertheless supplies a reusable diagnostic for other gapless bosonic lattice models: odd density–phase couplings should first be tested for an anticommutator representation and a ground-state-transform estimate, while direct even corrections should be tested for total commutators before they are bounded in operator norm.

A Coefficient audit

For convenience, we collect the normalization used throughout the paper:

$$H_{\bar{n}} = \frac{U}{2}\sum_x M_x^2 + J\sum_e (A_x - A_y)^\dagger (A_x - A_y), \tag{43}$$

$$H_0 = \frac{U}{2}\sum_x M_x^2 + J\sum_e \mathcal{A}_e, \qquad \mathcal{A}_e = 2 - T_e - T_e^\dagger, \tag{44}$$

$$D_{\mathrm{odd}} = \frac{J}{2}\sum_e q_e \mathcal{A}_e, \qquad b_x = \frac{J}{2}\sum_{e\ni x} \mathcal{A}_e, \tag{45}$$

$$H_2 = \frac{J}{8}\sum_e \left[T_e(d_e+1)^2 + (d_e+1)^2 T_e^\dagger\right], \tag{46}$$

$$H_2 = -\frac{1}{8U^2}[H_0,[H_0,V]] + \frac{1}{8U}|\nabla_\theta V|^2, \tag{47}$$

$$W_{\mathrm{sc}} = -\frac{J^2}{4U}\sum_x (Q_x^2 + \overline{Q}_x^2), \qquad J_* = J - \frac{qJ^2}{U\bar{n}^2}. \tag{48}$$

Changing the coefficient of the phase-gradient term from J to $J/2$ requires a global replacement of J in every hopping, force-square, and stiffness-renormalization formula. Mixing the two conventions produces the factor-of-two inconsistencies that motivated the normalization audit underlying this version.

B Statements not claimed

For clarity, this paper does not claim any of the following:

- a new thermodynamic Bose–Einstein condensation exponent or density regime;
- volume-uniform Gaussian domination for the original high-filling amplitude model;
- removal of the finite number window in the second-order operator-norm expansion;
- an all-order ferromagnetic effective Hamiltonian;
- identification of the complete fourth-order quantum normal form.

References

[1] Jakob E. Björnberg and Daniel Ueltschi. Reflection positivity and infrared bounds for quantum spin systems. In *50 Years of the Berezinskii–Kosterlitz–Thouless Theory*. EMS Press, 2023.

[2] Alexander Bols and Wojciech De Roeck. Asymptotic localization in the bose–hubbard model. *Communications in Mathematical Physics*, 358:1015–1070, 2018.

[3] Matthew P. A. Fisher, Peter B. Weichman, G. Grinstein, and Daniel S. Fisher. Boson localization and the superfluid-insulator transition. *Physical Review B*, 40:546–570, 1989.

[4] Jürg Fröhlich and Y. M. Park. Correlation inequalities and the thermodynamic limit for classical and quantum continuous systems. *Communications in Mathematical Physics*, 59:235–266, 1978.

[5] Jean Ginibre. General formulation of griffiths' inequalities. *Communications in Mathematical Physics*, 16:310–328, 1970.

[6] Tadahiro Miyao. Reflection positivity and quantum griffiths' inequalities, 2015.

[7] Tadahiro Miyao. Quantum griffiths inequalities. *Journal of Statistical Physics*, 164:255–303, 2016.

[8] T. P. Polak and T. K. Kopeć. Quantum rotor description of the mott-insulator transition in the bose–hubbard model. *Physical Review B*, 76:094503, 2007.

[9] Niklas Teichmann, Dennis Hinrichs, Martin Holthaus, and André Eckardt. Bose–hubbard phase diagram with arbitrary integer filling. *Physical Review B*, 79:100503(R), 2009.